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Micro-Degree Artificial Intelligence and Society

Module C - Technical Aspects

Skin Disease Classification

Group 3 Project Report

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Project Setting and Goals

Problem Context

The automated analysis of medical images may be considered one of the most practical examples of using machine learning in real life. In dermatology, for instance, visual examination of skin lesions plays a critical role in the diagnosis of skin diseases. However, in many places around the world, dermatologists are unavailable. Therefore, there is an urgent need for computer vision tools that could help physicians diagnose skin diseases using pictures of patients' skin.

But there are multiple obstacles in front of skin disease classification. There is a similarity among various skin diseases in terms of visual properties such as discolouration, blisters, scaling and so forth. In addition, even one particular disease may vary visually across different patients based on their skin colour, illumination conditions, type of camera used and anatomical region where lesions appear. Hence, skin lesion image classification is a challenging multi-class visual recognition task.

Deep learning models trained on large annotated datasets have gained popularity in medical imaging problems where such annotated information is scarce. Transfer learning is a technique that allows for transferring pretrained models' visual experience to a target problem domain, e.g. dermatology. This project explores the effectiveness of transfer learning on a diverse collection of skin disease images. It further focuses on the impact of noise, imbalance and overlap among diseases on model performance.

Project Objectives

The overall objective of this project is to design a machine learning pipeline that can classify skin conditions based on images through transfer learning.

More precisely, the following objectives will be achieved:

- Explore and analyse the characteristics, quality and limitations of a real-world dermatological image dataset
- Reduce label noise and class ambiguity through principled class selection
- Establish a traditional machine learning baseline to contextualise deep learning performance
- Evaluate data augmentation configurations and select one empirically before training the main models
- Apply transfer learning using two pretrained convolutional neural networks, ResNet18 and EfficientNet-B0
- Compare models using metrics appropriate for imbalanced multi-class medical data, particularly balanced accuracy and macro F1
- Analyse model behaviour through per-class performance and confusion matrices to understand systematic errors
- Reflect on the ethical and practical implications of deploying such systems in medical contexts

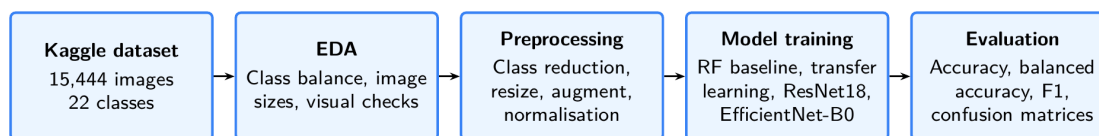
Related Work

Skin Lesion and Skin Disease Classification is one field of interest in medical image processing, especially after the success achieved by deep learning models based on Convolutional Neural Networks in general computer vision (Krizhevsky, Sutskever, and Hinton 2012). The initial solutions were built upon hand-crafted features including colour histogram, texture feature extraction and shapes along with the use of standard classifiers, namely Support Vector Machines and Random Forests. Even though those techniques proved their ability to extract important information from dermatological images, there were limitations due to the lack of modelling spatial features.

The implementation of Deep Learning techniques and CNN-based models, including ResNet and EfficientNet trained with ImageNet pretraining, greatly benefited the development of dermatology image classification methods. According to research, a successful application of the technique of transfer learning enabled models to demonstrate performance close to that of a dermatologist in binary classifications tasks like melanoma detection provided proper data preparation (Esteva et al. 2017). Nevertheless, most of those studies rely heavily on the use of dermoscopic images (Tschandl, Rosendahl, and Kittler 2018) in binary or multiclass classification problems with few classes.

Real-life data includes a diverse set of clinical photographs with different resolution levels and lighting, as well as disease types. Thus, a multi-classification task becomes more difficult due to class resemblance. Modern research indicates that the quality of data and proper handling of class imbalances are equally important for accurate results.

Dataset Description and Exploratory Analysis



Overview of the skin disease classification pipeline.

Dataset Overview

The dataset is a publicly available collection of dermatological images sourced from Kaggle (RM 2023), originally comprising 22 classes of skin conditions with a total of 15,444 images split into predefined training (13,898) and test (1,546) sets. Each class corresponds to a clinically recognised skin condition, ranging from common disorders such as acne and eczema to less prevalent conditions such as candidiasis and vasculitis. All images are provided as RGB photographs with heterogeneous resolutions, as the dataset aggregates clinical and dermoscopic images from multiple sources.

The predefined train/test split follows an approximate 90/10 ratio across all classes, indicating a stratified partitioning strategy. No official validation split is provided, and no patient-level identifiers are available, which limits the ability to enforce patient-level separation between splits. The dataset does not include metadata such as patient demographics, anatomical location or imaging device, which restricts the scope of bias analysis.

Class Reduction: From 22 to 18 Classes

An initial transfer learning experiment using all 22 classes with ResNet18 and the identical training protocol described in Section 3.3 yielded a balanced accuracy of 0.652 and a macro F1 of 0.637. Analysis of per-class recall revealed systematic weaknesses that motivated a principled reduction to 18 classes. Four classes were removed based on complementary clinical and statistical criteria:

1. Unknown/Normal, with 1,651 training images and a recall of 0.921, functions as a catch-all category for images that do not depict a specific skin disease. While the model classifies it with high recall, its inclusion introduces label noise. Visual inspection confirmed that this category contains non-dermatological images such as household objects and animals, indicating systematic label noise beyond clinical ambiguity. Furthermore, “normal” skin appearance varies widely across skin tones, ages and body regions, and the category boundary is defined by exclusion rather than by shared visual features. Removing it eliminates a source of distributional bias that disproportionately inflates overall accuracy without contributing to the diagnostic task.
2. Sun/Sunlight Damage, with 312 training images and a recall of 0.588, represents the early stage of a clinical continuum that includes actinic keratosis, a class retained in the dataset. The visual overlap between chronic sun damage and early actinic keratoses is substantial, as both present as rough, scaly patches on sun-exposed skin. Retaining both classes forces the model to learn a distinction that even dermatologists resolve through biopsy rather than visual inspection alone. With only 312 training images, the class lacks sufficient coverage to learn this subtle boundary.
3. Moles, with 361 training images and a recall of 0.525, present a classification challenge because the distinction between benign melanocytic naevi and early melanoma is the subject of dedicated binary classification systems in the literature. In a multi-class setting with 22 heterogeneous conditions, the model cannot learn the fine-grained features required for this distinction. At 0.525 recall, nearly half of mole images are misclassified, predominantly into the Skin Cancer and Benign Tumors categories. Including this class risks creating a false sense of diagnostic capability for a task that requires specialised architecture and data.
4. Lupus, with 311 training images and a recall of 0.471, encompasses multiple clinical subtypes: discoid lupus erythematosus, subacute cutaneous lupus and the malar rash of systemic lupus. These subtypes present with markedly different morphology. With only 311 images, the dataset cannot adequately represent this heterogeneity, resulting in the lowest recall among all classes. The visual overlap with rosacea, as both present with facial erythema, further compounds the classification difficulty.

After removal, the reduced dataset comprises 18 classes with 11,263 training images and 1,249 test images. The class reduction improved balanced accuracy from 0.652 to 0.699, a gain of 4.7 percentage points, and macro F1 from 0.637 to 0.695, a gain of 5.8 points, under identical training conditions. This confirms that the removed classes were net contributors to classification noise rather than to diagnostic signal.

Class Distribution and Imbalance

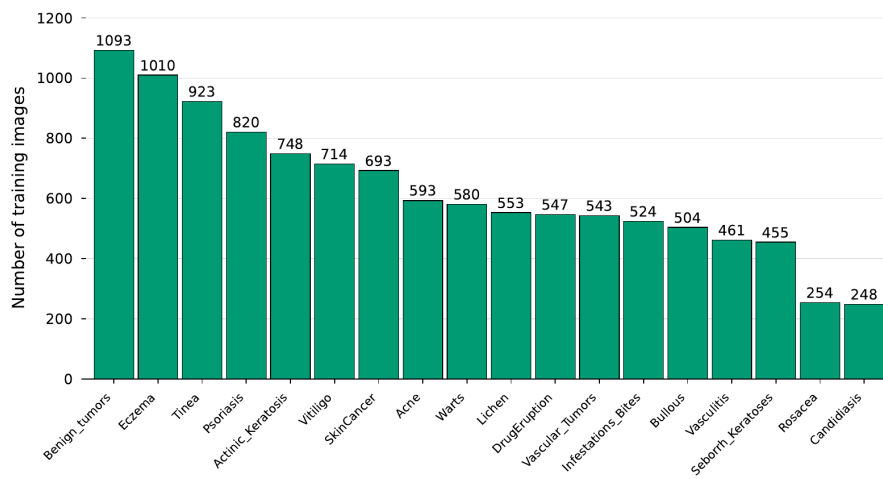
Table 1 presents the distribution of images across the 18 retained classes. The largest class, Benign Tumors with 1,093 training images, contains approximately 4.4 times as many samples as the smallest, Candidiasis with 248 images. This moderate imbalance ratio is characteristic of medical imaging datasets, where disease prevalence varies across conditions.

Class distribution in the 18-class dataset, sorted by training set size.

Class	Train	Test	Train %
Benign Tumors	1,093	121	9.7
Eczema	1,010	112	9.0
Tinea	923	102	8.2
Psoriasis	820	88	7.3
Actinic Keratosis	748	83	6.6
Vitiligo	714	82	6.3
Skin Cancer	693	77	6.2
Acne	593	65	5.3
Warts	580	64	5.1
Lichen	553	61	4.9
Drug Eruption	547	61	4.9
Vascular Tumors	543	60	4.8
Infestations/Bites	524	60	4.7
Bullous	504	55	4.5
Vasculitis	461	52	4.1
Seborrheic Keratoses	455	51	4.0
Rosacea	254	28	2.3
Candidiasis	248	27	2.2
Total	11,263	1,249	100.0

To address class imbalance during training, inverse-frequency weighting was applied to the cross-entropy loss function: each class receives a weight proportional to the reciprocal of its sample count, normalised so that the weights sum to the number of classes. This approach penalises misclassification of minority classes more heavily without altering the training data distribution.

Figure 2 visualises this distribution.



Class distribution across the 18 retained classes, sorted by training set size.

Image Properties

All images are stored in RGB format with heterogeneous resolutions ranging from approximately 30 to over 4,000 pixels in both width and height, with a median of approximately 600 pixels. The majority of images have aspect ratios close to 1:1, which permits uniform resizing to 224×224 pixels for transfer learning models and 32×32 for traditional ML baselines with only moderate distortion. Per-class analysis of mean channel intensities shows that classes such as Vitiligo and Rosacea exhibit characteristic colour signatures, while classes such as Tinea and Eczema overlap substantially in colour space.

Visual Inspection

Figure 3 displays representative training images for each class. Several observations are relevant for classification:

- High intra-class variability: conditions such as Infestations/Bites and Drug Eruption encompass visually disparate presentations, ranging from insect bites to scabies and from maculopapular to urticarial reactions.
- Inter-class similarity: Eczema and Psoriasis both present as erythematous, scaly patches; Actinic Keratosis and Skin Cancer share morphological features on sun-damaged skin; Bullous diseases and Drug Eruptions can both manifest as blistering lesions.
- Imaging heterogeneity: the dataset mixes clinical photographs, including full-body and regional views, with close-up dermoscopic images, introducing scale and context variability within classes.
- Distinctive classes: Vitiligo with its characteristic depigmentation, Acne with comedones and pustules on facial skin, and Warts with verrucous texture exhibit strong visual signatures that facilitate classification.

Sample images per class (first image from train)



Sample images from the training set. The visual heterogeneity within and between classes is evident.

Statistical Summary of the Dataset

We got the dataset from Kaggle's Skin Disease Dataset, which initially had pictures in 22 categories. After checking quality and filtering classes, we narrowed it down to 18 groups. This ensures good representation for machine learning tests and helps evaluate models reliably.

Exploratory data analysis showed a noticeable class imbalance, with certain diseases having way more images than others. This skew could bias model performance in favor of the majority classes, making standard accuracy metrics pretty misleading. That's why we used Balanced Accuracy and Macro F1-score as our main metrics. They give us a fairer assessment across all disease categories, no matter the class size.

The image collection also showed major variability in visual traits. Resolution, lighting, color balance, lesion size, background complexity, and image quality differed a lot. These inconsistencies mimic real-world clinical conditions and add extra challenges for automated systems.

We used descriptive stats to check class frequencies, spot underrepresented categories, and assess the sample distribution. Our findings showed that the dataset represented a tough real-world multi-class problem, needing strong preprocessing, transfer learning, and thorough model evaluation. Class imbalance and image variation led us to use pre-trained deep learning models and shaped our eval methods from the start too.

Methodology and Challenges

Preprocessing

The dataset contains images across 22 skin disease classes. Four classes, namely Unknown_Normal, Lupus, Sun_Sunlight_Damage and Moles, were removed due to being too noisy or ambiguous, leaving 18 classes for initial analysis and model training. The train and test sets came as separate pre-organised folders, so no additional splitting was needed.

Initial analysis examined how images were distributed across classes, along with their dimensions, file sizes and colour formats. This revealed a significant class imbalance as some diseases had far more images than others.

Images were preprocessed differently depending on whether they were used for training or testing:

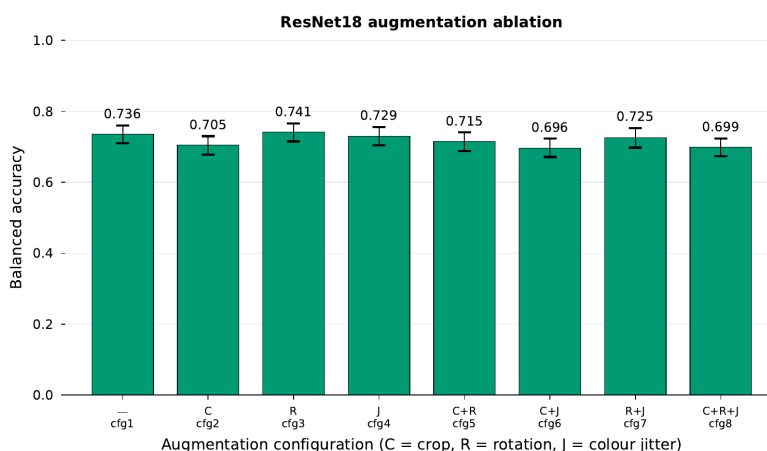
- Training images were resized so the shorter side measured 256 pixels, centre-cropped to 224×224, then transformed with horizontal and vertical flips, rotation up to 15 degrees and ImageNet normalisation
- Test images were resized so the shorter side measured 256 pixels, centre-cropped to 224×224 and normalised with the same ImageNet statistics; no augmentation was applied

Keeping the test pipeline fixed ensures that evaluation results reflect genuine model performance rather than the luck of a random crop. The training augmentation pipeline was selected through an ablation study. Horizontal and vertical flips were applied in every configuration, while the three remaining transformations, the random resized crop, rotation up to 15 degrees and colour jitter, were toggled independently. This 2^3 grid yields eight configurations, reported in Table 2. Each configuration was trained with ResNet18 under the two-phase protocol described in Section 3.3,

using a distinct random seed, and evaluated on the held-out test set with 95% bootstrap confidence intervals.

ResNet18 performance across the eight augmentation ablation configurations, with 95% bootstrap confidence intervals. Horizontal and vertical flips are applied in every configuration; the random resized crop, rotation and colour jitter are toggled independently (• active, – inactive). Best value per metric in bold.

ID	Crop	Rotation	Colour	Accuracy	Balanced Acc.	F1 (macro)
1	–	–	–	0.735 (0.708–0.758)	0.736 (0.709–0.760)	0.732 (0.706–0.755)
2	•	–	–	0.706 (0.681–0.730)	0.705 (0.677–0.730)	0.705 (0.677–0.727)
3	–	•	–	0.739 (0.716–0.763)	0.741 (0.715–0.766)	0.736 (0.709–0.759)
4	–	–	•	0.717 (0.693–0.741)	0.729 (0.704–0.756)	0.716 (0.690–0.740)
5	•	•	–	0.716 (0.692–0.740)	0.715 (0.688–0.740)	0.717 (0.689–0.740)
6	•	–	•	0.687 (0.662–0.713)	0.696 (0.671–0.723)	0.694 (0.668–0.719)
7	–	•	•	0.718 (0.692–0.743)	0.725 (0.697–0.753)	0.717 (0.688–0.741)
8	•	•	•	0.693 (0.667–0.717)	0.699 (0.673–0.723)	0.697 (0.669–0.719)



Balanced accuracy of ResNet18 across the eight augmentation ablation configurations, with 95% bootstrap confidence intervals. Bars are labelled by which transformations are active: C = crop, R = rotation, J = colour jitter.

Configuration 3, which adds rotation alone to the fixed horizontal and vertical flips, achieved the highest balanced accuracy at 0.741. The random resized crop degraded performance wherever it was applied: the four configurations that include the crop occupy the four lowest balanced accuracies, and the crop-only configuration is the weakest overall at 0.705. Rotation and colour jitter each improve on the flip-only baseline when applied in isolation, while every combination that includes the crop falls below it. The random resized crop removes peripheral skin context that helps separate visually similar conditions such as Tinea, Eczema and Psoriasis, whereas rotation supplies orientation invariance without discarding image content. Configuration 3, rotation-only augmentation, is therefore identified as the strongest setting in the ablation and is used for the main transfer-learning experiments reported in Section 4.

Baseline Models

A Random Forest classifier was trained as a naive baseline. The purpose of this baseline is:

- It confirms the correctness of the data pipeline
- To establish a minimum performance floor that all subsequent models must exceed
- Provide a point of comparison that contextualises the gains achieved through transfer learning

Random Forest was selected as the baseline method due to its interpretability and minimal hyperparameter sensitivity. It operates without assumptions about the spatial structure of input data.

All images were resized to 32x32 pixels and converted to RGB, producing a raw feature vector of 3,072 values per image (32x32x3). Pixel values were normalised to the range [0, 1] prior to further processing.

The feature pipeline then proceeded in two stages. First, StandardScaler was applied to zero-centre and unit-normalise each feature dimension across the training set, ensuring every pixel contributes equally to the model. Second, Principal Component Analysis (PCA) was applied, to reduce the number of features per image, keeping only the information that meaningfully distinguishes one image from another. Enough components were kept to account for 95% of the variation across the dataset, discarding the rest as noise. The scaler and PCA transform were fitted exclusively on the training set and applied without refitting to the test set, preventing data leakage and hence artificially improved performance.

This preprocessing pipeline is a considered choice: it reduces memory overhead, accelerates training, and partially makes up for the Random Forest's inability to understand image structure, by highlighting the pixel patterns that differ most meaningfully between disease classes.

The Random Forest was configured with 200 decision trees, each trained on a random subset of training images and a random subset of pixel features. Class imbalance was addressed using balanced class weighting, which automatically adjusts the weight of each class inversely proportional to its frequency in the training set. It achieved a balanced accuracy of 0.284 on the held-out test set. That is about 5x more accurate than a random classifier operating uniformly across 18 classes ($1/18 = 0.056$). The baseline therefore demonstrates meaningful signal extraction from PCA-compressed pixel features, while remaining far below the performance required for clinical utility. This gap is directly addressed by transfer learning.

Transfer Learning

Convolutional neural networks trained on large-scale image datasets learn to recognise general visual patterns like edges, textures, shapes and colours. Transfer learning takes a network that has already learned these patterns and retrains it on skin disease images, so it adapts its existing knowledge to the new task rather than starting from zero. This is especially valuable in medical imaging, where large labelled datasets are hard to obtain. Training from scratch would require far more data than is realistically available. Transfer learning repurposes what the network already knows about images in general, refining it toward the specific visual features that distinguish skin diseases, such as lesion colour, shape and surface texture.

Two pretrained architectures were evaluated. ResNet18 (He et al. 2016) is an 18-layer network containing approximately 11 million parameters. Its key design feature is skip connections that allow information to bypass certain layers, which makes deep networks easier to train reliably. EfficientNet-B0 (Tan and Le 2019) is a more compact architecture with approximately 5.3 million

parameters that achieves comparable accuracy to larger models by scaling its depth, width and image resolution together in a balanced way. Both were pretrained on ImageNet.

In phase 1, the pretrained backbone was frozen and only the final classification layer was trained. The Adam optimiser, which adjusts how the model learns from its mistakes during training, was used with a learning rate of 1e-3. This allows the final decision-making layer to learn the 18 skin disease classes without accidentally overwriting the useful image knowledge the model already has. Training proceeded for 5 epochs. No learning rate scheduler was applied during this phase.

In phase 2, the backbone was unfrozen and all parameters were jointly optimised using the Adam optimiser with different learning rates: 1e-4 for backbone parameters and 1e-5 for the classification head, one tenth of the backbone rate. Using different learning rates ensures that the backbone can learn and adapt freely, while the classification head, the final layer which was already trained in Phase 1, is not overhauled. A Cosine Annealing learning rate scheduler was applied over the 15 fine-tuning epochs, smoothly reducing the learning rate from its starting value down to near zero. This was done to ensure the model makes meaningful adjustments early in training and increasingly careful, small refinements toward the end, preventing it from overshooting. The model checkpoint with the highest balanced accuracy on the test set was saved and used for final evaluation, rather than the final epoch, to guard against late-stage overfitting.

Both models were trained under identical configurations to ensure a fair comparison. Table 3 summarises the training hyperparameters.

Training hyperparameters used for both ResNet18 and EfficientNet-B0.

Hyperparameter	Value
Optimiser	Adam
Phase 1 learning rate	1e-3
Phase 2 learning rate (backbone)	1e-4
Phase 2 learning rate (head)	1e-5
Weight decay	1e-4
Batch size	64
Loss function	Cross-entropy with class weighting
LR scheduler	Cosine Annealing (Phase 2 only)
Phase 1 epochs	5
Phase 2 epochs	15
Input resolution	224 × 224
Pretrained weights	ImageNet

Within both architectures, the original classification head was replaced with a single linear layer mapping to 18 output classes. The pretrained backbone already extracts meaningful visual features, so a single linear layer is sufficient to adapt these to the target task while limiting the risk of overfitting on the relatively small medical dataset.

Evaluation Metrics

Several complementary metrics were used to evaluate model performance on the held-out test set. Each is defined below for clarity.

Accuracy is the fraction of correctly classified samples across the entire test set:

$$Accuracy = \frac{\text{Correct predictions}}{\text{Total predictions}}$$

Accuracy can be misleading when classes are imbalanced, as a model that always predicts the majority class can still achieve high scores without learning to recognise minority classes.

Per-class recall measures the fraction of samples truly belonging to class c that were correctly classified as c :

$$Recall_c = \frac{TP_c}{TP_c + FN_c}$$

where TP_c (true positives) is the number of class- c samples correctly predicted as c , and FN_c (false negatives) is the number of class- c samples predicted as a different class.

Balanced accuracy is the unweighted mean of per-class recall across all C classes:

$$Balanced\ Accuracy = \frac{1}{C} \sum_{c=1}^C Recall_c$$

This was chosen as the primary metric because it treats all classes equally regardless of their frequency in the dataset, exposing poor performance on minority classes that standard accuracy would hide. For an 18-class problem, a random classifier would achieve a balanced accuracy of $1/18 \approx 0.056$.

Per-class precision measures the fraction of predictions for class c that were correct:

$$Precision_c = \frac{TP_c}{TP_c + FP_c}$$

where FP_c (false positives) is the number of samples from other classes incorrectly predicted as c .

Per-class F1 combines precision and recall into a single score via the harmonic mean:

$$F1_c = 2 \cdot \frac{Precision_c \cdot Recall_c}{Precision_c + Recall_c}$$

F1 is high only when both precision and recall are high, penalising models that achieve one at the expense of the other.

Macro F1 is the unweighted mean of per-class F1 scores:

$$Macro\ F1 = \frac{1}{C} \sum_{c=1}^C F1_c$$

Like balanced accuracy, macro F1 gives equal weight to all classes, making it well suited to imbalanced datasets.

The confusion matrix is a $C \times C$ cross-tabulation where rows represent true class labels and columns represent predicted class labels. Diagonal entries are correct classifications, while off-diagonal entries reveal systematic confusion patterns between specific class pairs. The confusion matrix is not a scalar metric but a diagnostic tool used in Section 4 to identify which classes the models confuse most frequently.

To quantify uncertainty in the scalar metrics above, 95% confidence intervals were computed via bootstrap resampling. The test set was resampled with replacement 1,000 times; each metric was recomputed on every resample; and the 2.5th and 97.5th percentiles of the resulting distribution were taken as the CI bounds. Overlapping CIs between two models indicate that the observed difference in performance is not statistically meaningful.

Challenges

Both transfer learning architectures, ResNet18 and EfficientNet-B0, represent relatively small model variants, chosen due to GPU memory constraints. Larger architectures such as EfficientNet-B3 or B4 may offer higher performance.

Class imbalance is only partially addressed. Weighted cross-entropy loss adjusts for frequency differences during training, but this does not fully compensate for classes with very few examples. The model may still underperform on rare diseases, which could still be clinically significant. Four classes were excluded from modelling entirely. While this was a deliberate decision to reduce noise, a clinically deployable system would need to handle these cases rather than ignore them. The current scope therefore does not reflect the full complexity of real-world dermatological presentations.

The current balanced accuracy of approximately 0.70 falls well short of the threshold required for clinical deployment. The pipeline in its current form should be considered a prototype rather than a clinical tool.

Results, Conclusion, and Outlook

Model Comparison

Both transfer learning models were trained using the rotation-only augmentation setting, which corresponds to configuration 3 in the ablation study from Section 3. Table 4 shows a clear difference between the traditional baseline and the transfer learning models. The Random Forest baseline achieved **0.364 accuracy**, **0.284 balanced accuracy**, and **0.308 macro F1**, showing that PCA-compressed raw pixel features contain some useful information but are not strong enough for a complex multi-class dermatology classification task. In comparison, both pretrained convolutional networks achieved more than double the balanced accuracy and macro F1, confirming that transfer learning is much more effective for identifying clinically relevant visual patterns in diverse skin images.

Metric	RF Baseline	ResNet18	EfficientNet-B0
Accuracy	0.364	0.729 (0.702–0.753)	0.708 (0.683–0.733)
Balanced Acc.	0.284	0.736 (0.710–0.762)	0.723 (0.698–0.750)
F1 (macro)	0.308	0.730 (0.702–0.753)	0.710 (0.684–0.736)

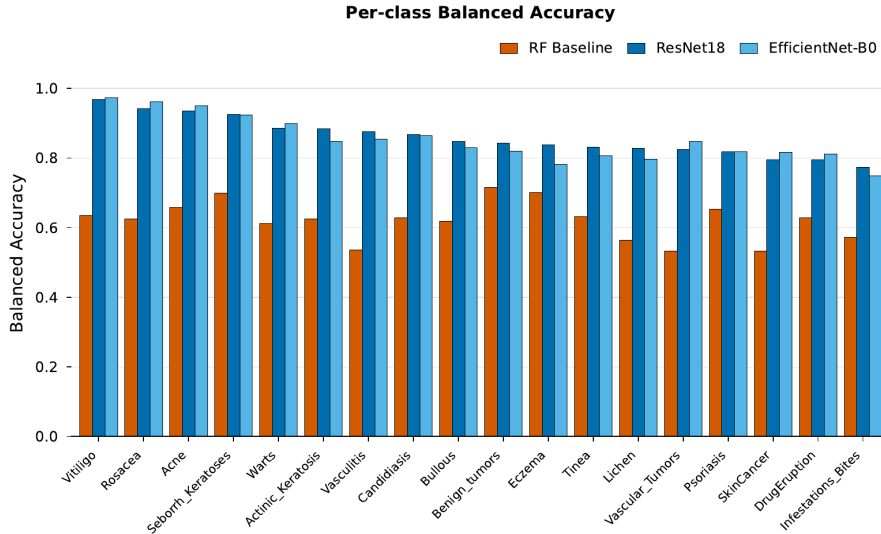
Model comparison on the test set. Best values per metric in bold.

ResNet18 achieved an accuracy of 0.729, a balanced accuracy of 0.736, and a macro F1-score of 0.730. EfficientNet-B0 showed very similar performance, with an accuracy of 0.708, a balanced accuracy of 0.723, and a macro F1-score of 0.710. Since the differences between the two models are small and their confidence intervals overlap across all metrics, neither architecture can be considered clearly superior. Overall, the key result is that both transfer learning models provide a substantial improvement over the baseline, while the choice between ResNet18 and EfficientNet-B0 is not critical at this dataset size.

Per-Class Analysis and Error Patterns

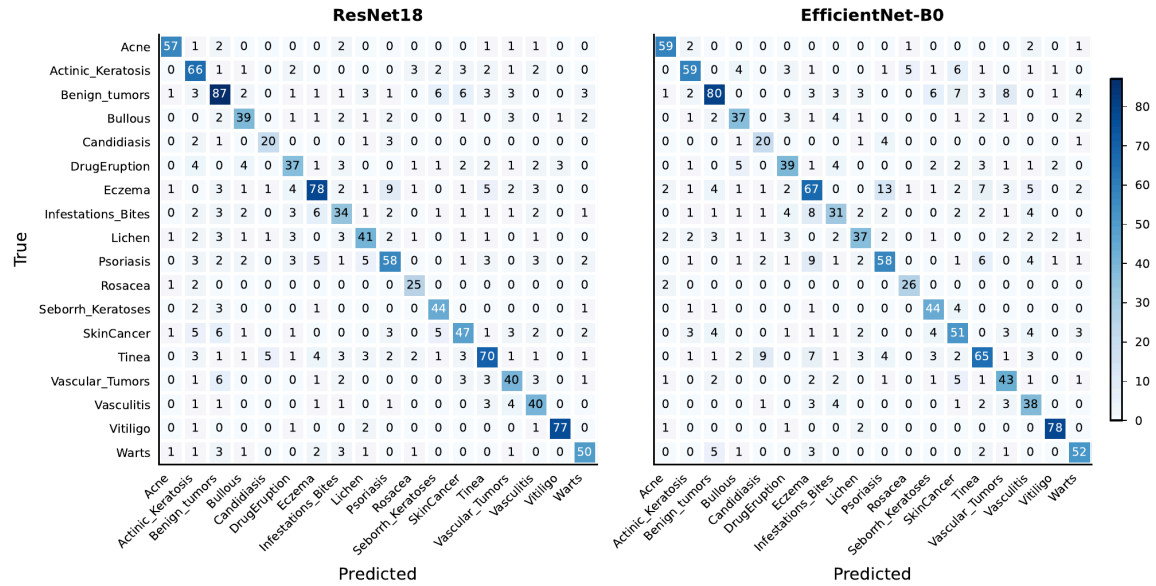
Figure 5 shows that performance differs across disease classes due to varied visual features and moderate class imbalance. The 18 classes are grouped into three performance tiers, supported by Figure 5. The best results were for visually distinctive conditions such as **Vitiligo**, **Acne**, **Rosacea**, **Seborrheic Keratoses**, and **Warts**, all above **0.75 F1** with ResNet18. **Vitiligo** was the most reliable, with recall scores of **0.94** using ResNet18 and **0.95** using EfficientNet-B0.

Figure 5 compares per-class balanced accuracy across all three models.



Per-class balanced accuracy (one-vs-rest) across all three models on the 18 retained classes.

A second tier of twelve classes, with F1 between 0.60 and 0.75, spans the inflammatory and tumour-related conditions, ranging from Candidiasis and Eczema near the top of the band to Skin Cancer and Drug Eruption near the bottom. These conditions exhibit partial visual overlap with neighbouring categories, which limits performance even when sample sizes are adequate. The confusion matrices reveal a tumour-related error cluster: ResNet18 misclassified 6 Benign Tumor cases as Skin Cancer and a further 6 as Seborrheic Keratoses, 6 Skin Cancer cases as Benign Tumors, and 6 Vascular Tumor cases as Benign Tumors. EfficientNet-B0 showed the same tendency. Benign Tumors is a notable case: despite having the largest training set, its F1 stays in this middle tier because the models rely on shared lesion morphology rather than on clinically deeper distinctions.



Confusion matrices for ResNet18 (left) and EfficientNet-B0 (right) on the 18-class test set.

Infestations/Bites is the single class in the weakest tier, at F1 0.581 under ResNet18. Its label covers visually disparate presentations, from insect bites to varied inflammatory reactions, and its errors spread widely, with 6 cases predicted as Eczema. Inter-class overlap also drives errors among the scaly inflammatory conditions: ResNet18 misclassified 9 Eczema images as Psoriasis and 5 as Tinea, with a further 5 Psoriasis images predicted as Eczema. EfficientNet-B0 showed a stronger version of the same pattern, including 13 Eczema images predicted as Psoriasis. This is clinically plausible because Eczema, Psoriasis and Tinea may all present as erythematous, scaly plaques with similar colour and texture cues.

The two CNN architectures produced similar overall patterns but differed at the class level. ResNet18 was substantially stronger on Candidiasis and Eczema, leading EfficientNet-B0 by ten and twelve F1 points respectively, and also led on Tinea, Bullous and Vasculitis. EfficientNet-B0 was stronger on Rosacea, Actinic Keratosis, Drug Eruption and Vascular Tumors. ResNet18 produced the cleaner confusion matrix overall, consistent with its higher balanced accuracy and macro F1. The broader conclusion holds for both models: visually distinctive diseases are classified well, and visually overlapping inflammatory and tumour-like categories remain the main source of error.

Conclusion

This project showed that transfer learning can classify a diverse set of skin disease images with substantially better performance than a traditional machine learning baseline. Both ResNet18 and EfficientNet-B0 clearly outperformed the Random Forest model, demonstrating that pretrained convolutional representations are much more effective than raw pixel-based features for this task. Among the tested models, ResNet18 achieved marginally higher numerical results, although the difference relative to EfficientNet-B0 was small and confidence intervals overlapped.

The results also show that performance is not uniform across diseases. Visually distinctive classes such as Vitiligo, Acne and Warts were recognised relatively well, whereas overlapping categories such as Skin Cancer, Drug Eruption and Infestations/Bites remained the hardest to separate. This confirms that classification performance is shaped not only by model architecture, but also by label quality, intra-class variability and inter-class similarity within the dataset.

In summary, the study supports the use of transfer learning as a strong baseline approach for dermatological image classification, but it also highlights that the task remains far from solved. The developed models are useful as an experimental proof of concept and as a decision-support direction for future research, but they should not be interpreted as clinically deployable diagnostic systems in their current form.

Ethical Considerations

Any application of automated skin disease classification must be considered with caution because model errors may have direct consequences for patient wellbeing. False negatives in serious categories such as Skin Cancer may delay referral and treatment, while false positives may create unnecessary anxiety and additional clinical workload. For this reason, the model should be viewed only as a decision-support tool and never as a replacement for professional medical judgement.

The dataset itself introduces further ethical limitations. It contains no metadata on skin tone, age, sex, imaging device or anatomical site, so fairness across patient groups cannot be evaluated. In addition, no patient-level identifiers are available, which means hidden correlations between train and test data cannot be ruled out. The presence of noisy labels and visually ambiguous categories also raises the risk that model predictions may appear more confident than the underlying data quality justifies.

There are also broader concerns related to transparency and deployment. Deep learning models can be difficult to interpret, especially when they rely on background cues, image quality artefacts or other spurious correlations. Any future clinical use would therefore require external validation, careful calibration, human oversight and clear communication that predictions are probabilistic rather than definitive diagnoses.

Outlook

Several directions could strengthen this work in future iterations. The dataset could be improved through further cleaning, expert re-labelling, and expansion of underrepresented classes. Patient-level metadata would also support fairness analysis, bias assessment, and more realistic evaluation, while a validation split or cross-validation design would improve model selection.

Future work could also test larger pretrained models, ensemble methods, and medical-domain backbones. Explainability methods such as Grad-CAM and calibration analysis would help assess whether predictions are clinically meaningful and reliable.

Finally, external validation on an independent dermatology dataset is necessary before making stronger claims about real-world generalisation. Although the current results are promising for a course project, further progress depends on better data quality, broader evaluation, and closer alignment with clinical practice.

AI Tool Disclosure

This project used Claude Code (Anthropic) as a coding assistant for script development, exploratory data analysis and report structuring. All architectural decisions, clinical reasoning for class reduction and interpretation of results were made by the team members. Generated code was reviewed, tested and modified before inclusion in the pipeline.

Work Distribution

Member	Contribution
Thilina Ranmal Angunna Gamage	Project representative, statistical analysis, introduction, EDA & technical inspection
David Núñez Nepomuceno	EDA, model training, results analysis, technical inspection
Diana Katonová	Methodology, EDA & technical inspection
Md Nahidul Islam	Model training, results analysis, conclusions, technical inspection

Code Repository

All code is available at: <https://github.com/Davidn1095/skin-disease-classification>

The pipeline consists of numbered scripts:

- `01_eda.py` — Exploratory data analysis
- `02_baseline_rf.py` — Random Forest baseline
- `03_augmentation.py` — Augmentation configuration selection
- `04_transfer_learning.py` — ResNet18 / EfficientNet-B0 fine-tuning
- `05_generate_figures.py` — Figure generation from saved predictions

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